



It is yellow and transparent.

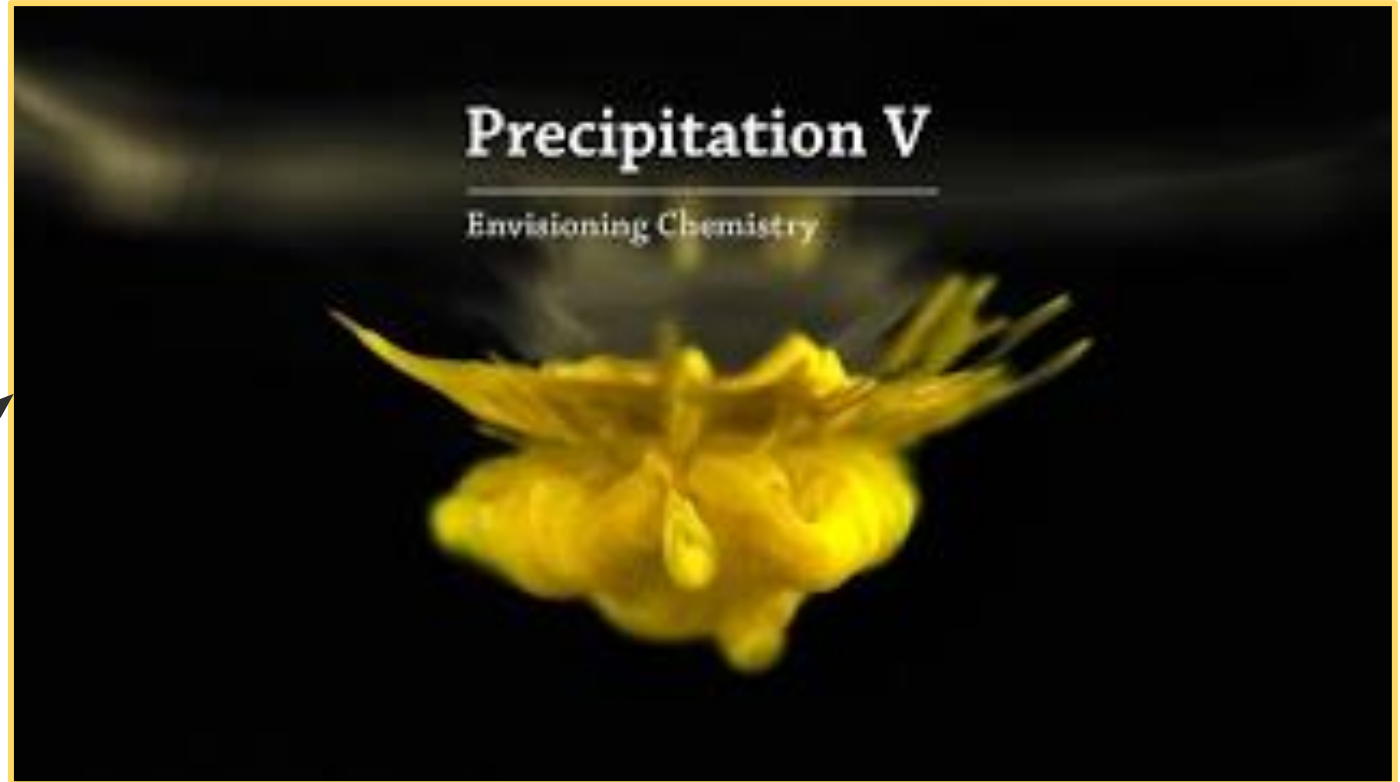
Precipitation Reactions

But now it's red and opaque and solid.



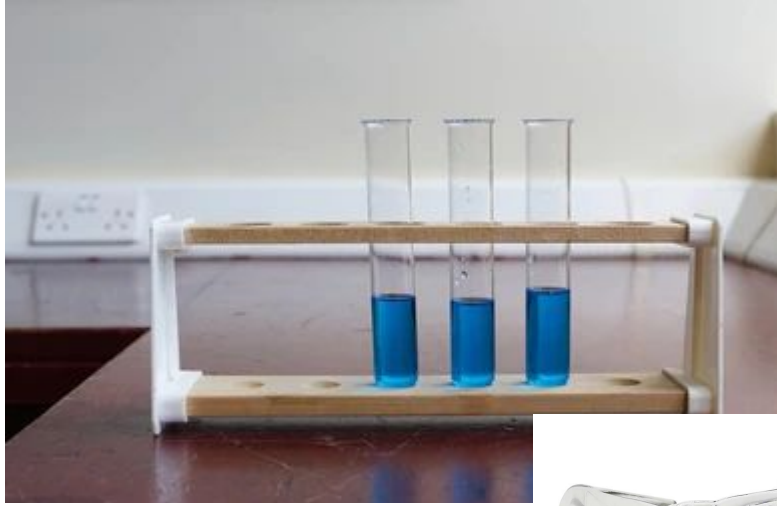
What actually is a precipitate?

- Precipitates are **solids** deposited in solutions.



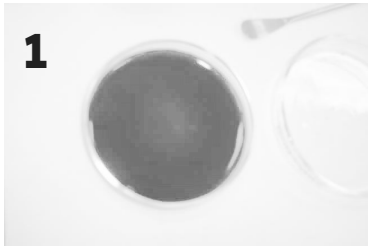
We did one last
week ;)

Are you equipped?

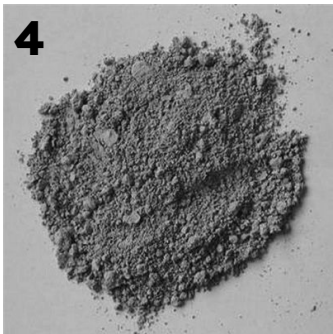


Method

Reactant No.1 <i>Goes in first.</i>	Reactant No.2 <i>Added w/pipette.</i>
Silver Nitrate (AgNO_3)	Potassium Chromate (K_2CrO_4)
Sodium Carbonate (Na_2CO_3)	Cobalt Chloride (CoCl_2)
Sodium Hydroxide (NaOH)	Cobalt Chloride (CoCl_2)
	Copper Sulfate (CuSO_4)
Iron Chloride (FeCl_3)	Potassium Thiocyanate (KSCN)



A. Copper Hydroxide
($\text{Cu}(\text{OH})_2$)



D. Cobalt Carbonate
(CoCO_3)



E. Iron Thiocyanate
($\text{Fe}(\text{SCN})_3$)



C. Silver Chromate
(Ag_2CrO_4)



B. Cobalt Hydroxide
($\text{Co}(\text{OH})_2$)

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D. Cobalt Carbonate
(CoCO_3)



C. Silver Chromate
(Ag_2CrO_4)



E. Iron Thiocyanate
($\text{Fe}(\text{SCN})_3$)

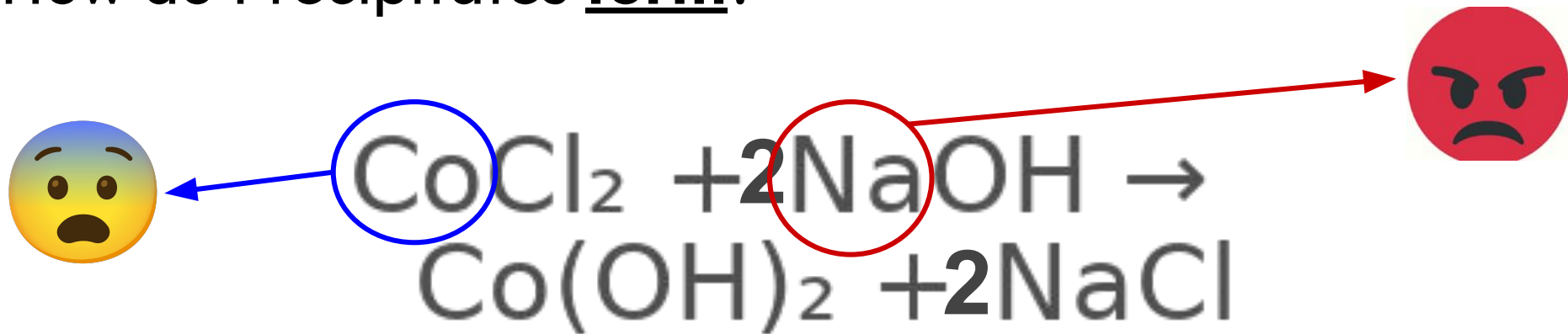


B. Cobalt Hydroxide
($\text{Co}(\text{OH})_2$)

Method

Reactant No.1 <i>Goes in first.</i>	Reactant No.2 <i>Added w/pipette.</i>	Precipitate formed?
Silver Nitrate (AgNO ₃)	Potassium Chromate (K ₂ CrO ₄)	C. Silver Chromate (Ag ₂ CrO ₄) 
Sodium Carbonate (Na ₂ CO ₃)	Cobalt Chloride (CoCl ₂)	D. Cobalt Carbonate (CoCO ₃) 
Sodium Hydroxide (NaOH)	Cobalt Chloride (CoCl ₂)	 B. Cobalt Hydroxide (Co(OH) ₂)
	Copper Sulfate (CuSO ₄)	A. Copper Hydroxide (Cu(OH) ₂) 

How do Precipitates form?



Sodium is very reactive, **more** reactive than Cobalt.

Sodium **displaces** Cobalt from its compound.

- They swap places.

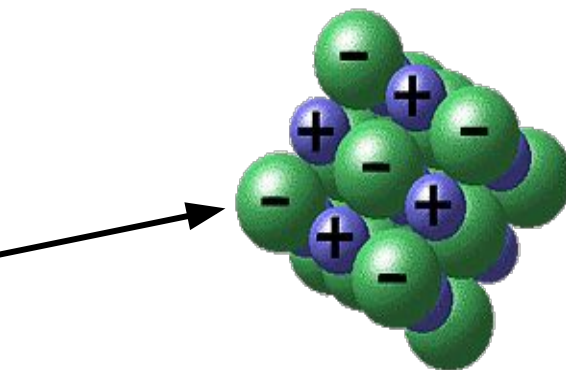
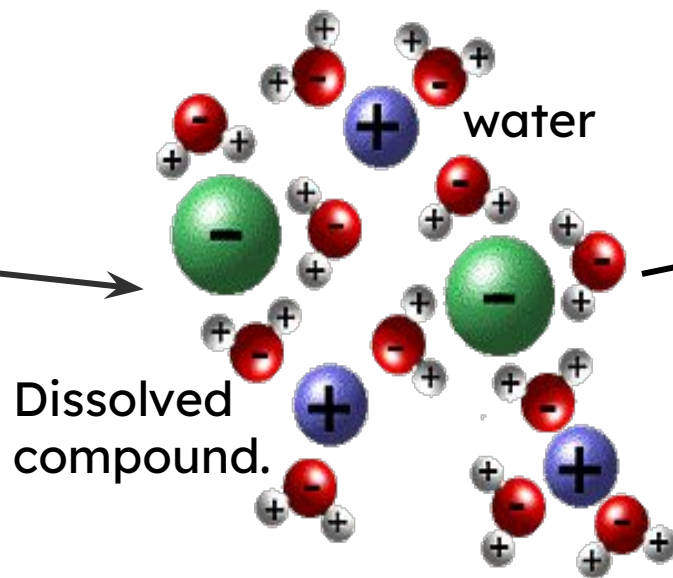
This is called a **displacement reaction**.

*Not strictly displacement -
but it's a secret*

How do Precipitates form?

All these compounds are **solid** at room temperature.

- They are just **dissolved** in water.



The substance we form however, **cannot dissolve.**

- This is the **precipitate.**